

Jaimin Suthar

jaiminsuthar390@gmail.com | +1(562) 254-2644 | linkedin.com/in/jaimin-suthar-251278222

Summary

Mechanical Engineering student (GPA: 3.57) with hands-on experience in **mechanical design, manufacturing, hardware testing, and system integration** across energy, aerospace, and electromechanical systems. Experienced in taking hardware from **CAD and engineering analysis through prototyping, fabrication, testing, and design iteration**, with strengths in FEA, thermal analysis, DFM, materials, and experimental validation. Proficient in SolidWorks, ANSYS, MATLAB, Python, LabVIEW, and engineering data acquisition, with experience collaborating across mechanical, electrical, controls, and materials teams.

Education

California State University, Long Beach

Aug 2022 – Dec 2026

- Bachelor's in Mechanical Engineering, Minor in Computer Science **GPA: 3.87/4.0**
- **Relevant Coursework:** Thermodynamics, Heat Transfer, Fluid Mechanics, Mechanical Design, Finite Element Analysis, Engineering Materials, Statistics for Engineers, Engineering Documentation

Experience

Reliability Engineering Intern, HyAxiom

May 2026 – Present

- Lead reliability and durability testing of **pneumatic control valves and balance-of-plant components** for PEM water electrolysis systems, developing test procedures, performance benchmarks, operational limits, and pass/fail criteria for hardware validation.
- Develop **LabVIEW-based data acquisition and testing workflows** using sensors and instrumentation to monitor component performance and capture operating conditions during hardware testing.
- Analyze sensor and historian data using **MATLAB, Python, signal processing, and RMS vibration analysis** to evaluate component behavior, identify abnormal operating conditions, and support engineering design decisions.

UAV Engineering Head, Beach Aerial Division

Oct 2025 – Present

- Led team of 25 independent researcher across several UAV platforms developing in the division. Designed, integrated and fabricated a **computer vision-assisted autonomous Quad-copter**, enabling mission execution including payload mechanism.
- Implemented computer vision pipeline using OpenCV and Python for target detection and navigation feedback. Also integrated **flight controllers, RC systems, and onboard electronics**, including wiring, power distribution, and sensor interfaces to enable reliable autonomous operation

Research Engineer, California State University – Long Beach, CA

Dec 2024 – Jan 2026

- Led mechanical design and fabrication of **3D-printed sensor housings and mechanical enclosures** using SolidWorks and additive manufacturing, optimizing geometry for fit, durability, and functional integration.
- Directed **cross-disciplinary collaboration** between mechanical, electrical, and materials sub-teams to prototype a **self-powered wearable system** integrating Triboelectric Nanogenerators (TENGs).

Projects

Thermal UAV - | Beach Launch Team

May 2023 – Sept 2023

- S Designed and fabricated a **3D-printed UAV airframe** using CAD and ASA material, developing the mechanical architecture around payload, sensor, flight-controller, and structural requirements.
- Performed **airflow and thermal analysis using ANSYS** to evaluate aerodynamic and thermal behavior, iterating the airframe design to improve airflow management and system performance. Developed a thermal-imaging system capable of **real-time fire detection and GPS-coordinate reporting**, integrating mechanical, sensing, and autonomous subsystems for field surveillance.

Brake Caliper Designer | FSAE Team

Jan 2024 – Feb 2025

- Designed and manufactured **mechanical braking systems** using SolidWorks and ANSYS, owning components from CAD through physical validation
- Improved braking system performance by **14%** and increased thermal dissipation efficiency by **28%** through iterative design and testing. Worked closely with electrical and vehicle dynamics teams to ensure system-level integration and reliability

Skills

- **Mechanical Design & Production** : SolidWorks, CATIA, AutoCAD, Engineering Drawings, Design for Manufacturability (DFM), Tolerance Analysis, Mechanical Assembly, Prototype Development
- **Analysis & Testing** : ANSYS, Abaqus, Finite Element Analysis (FEA), Structural Analysis, Thermal Analysis, Fluid/Airflow Analysis, MATLAB
- **Manufacturing & Materials:** Prototype Fabrication, Manufacturing Validation, Process Optimization
- **Programming & Controls:** Hardware Validation, Durability Testing, High-Cycle Testing, LabVIEW, Data Acquisition (DAQ)